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Inventor(s)	Liu; Yi-Ching et al.

Memory circuits

Abstract

A circuit is provided. The circuit includes a first die that includes a memory array, and the memory array includes a plurality of memory cells, a sensing element coupled to the plurality of memory cells, and a first plurality of conductive pads coupled to the sensing element. The circuit also includes a second die that includes an address decoder associated with the memory array of the first die and a second plurality of conductive pads coupled to the address decoder. The first die is coupled to the second die by an interposer. The address decoder of the second die is coupled to the sensing element of the first die. A first voltage swing of the first die is larger than a second voltage swing of the second die.

Inventors: Liu; Yi-Ching (Hsinchu, TW), Wang; Yih (Hsinchu, TW), Huang; Chia-En (Hsinchu County, TW)

Applicant: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY LTD. (Hsinchu, TW)

Family ID: 82930104

Assignee: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY LTD. (Hsinchu, TW)

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Primary Examiner: Hidalgo; Fernando

Attorney, Agent or Firm: WPAT LAW

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit of U.S. patent application Ser. No. 17/811,903 filed on Jul. 12, 2022, and U.S. patent application Ser. No. 17/225,913 filed on Apr. 8, 2021 (issued as U.S. Pat. No. 11,424,233), which are hereby incorporated herein by reference.

BACKGROUND

(2) A memory device, such as a semiconductor memory, is a data storage device, such as an

electronic data storage device. The memory device is often used as computer memory and implemented as a circuit, such as a semiconductor-based integrated circuit.

(3) Memory devices are made in many different types and technologies. Electrical memory has much faster access times than other types of data storage technologies. For example, a byte of data can often be written to or read from electrical memory within a few nanoseconds, while access times for magnetic storage, such as hard disks, is in the range of milliseconds. For these reasons, among others, an electric memory such as a semiconductor memory is often used as a primary storage mechanism for computer memory to hold data the computer is currently working on, among other uses.

(4) The subject matter discussed in the background section should not be assumed to be prior art merely as a result of its mention in the background section. Similarly, a problem mentioned in the background section or associated with the subject matter of the background section should not be assumed to have been previously recognized in the prior art. The subject matter in the background section merely represents different approaches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a circuit in accordance with some embodiments of the present disclosure.

(3) FIG. 2 illustrates a memory array in accordance with some embodiments of the present disclosure.

(4) FIG. 3 illustrates a circuit in accordance with some embodiments of the present disclosure.

(5) FIG. 4 illustrates a circuit in accordance with some embodiments of the present disclosure.

(6) FIG. 5 illustrates a circuit in accordance with some embodiments of the present disclosure.

(7) FIG. 6 illustrates a circuit in accordance with some embodiments of the present disclosure.

(8) FIG. 7 illustrates an exemplary flowchart in accordance with some embodiments of the present disclosure.

(9) FIG. 8 illustrates an exemplary flowchart in accordance with some embodiments of the present disclosure.

(10) FIG. 9 is a block diagram of an integrated circuit (IC) manufacturing system **1000**, and an IC manufacturing flow associated therewith, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) Other features and processes may also be included. For example, testing structures may be

included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(14) In the present disclosure, two layers/patterns/structures being formed on a same level means that the two layers/patterns/structures have a same distance from a reference plane, for example, a surface of a substrate, based on which a semiconductor device is formed, or the two layers/patterns/structures are intended to be configured to have a same distance from a reference plane, for example, a surface of a substrate, based on which a semiconductor device is formed but may not perfectly have the same distance from the reference plane due to design, manufacturing, measurement errors/margins caused by unperfected manufacturing and measurement conditions. Such a description should be recognizable to one of ordinary skill in the art.

(15) In the present disclosure, two layers/patterns/structures being formed on different level means that with consideration of variations/errors caused by, for example, surface roughness, the two layers/patterns/structures have different distances from a reference plane, for example, a surface of a substrate, based on which a semiconductor device is formed.

(16) In the present disclosure, a phrase “one of A, B and C” means “A, B and/or C” (A, B, C, A and B, A and C, B and C, or A, B and C), and does not mean one element from A, one element from B and one element from C, unless otherwise described.

(17) In the present disclosure, “about” or “approximately” used to describe a value of a parameter means that the parameter is equal to the described value or that the parameter is within a certain range of the described value, when design error/margin, manufacturing error/margin, measurement error etc. are considered. Such a description should be recognizable to one of ordinary skill in the art.

(18) Any of the embodiments described herein may be used alone or together with one another in any combination. The one or more implementations encompassed within this specification may also include embodiments that are only partially mentioned or alluded to or are not mentioned or alluded to at all in this brief summary or in the abstract. Although various embodiments may have been motivated by various deficiencies with the prior art, which may be discussed or alluded to in one or more places in the specification, the embodiments do not necessarily address any of these deficiencies. In other words, different embodiments may address different deficiencies that may be discussed in the specification. Some embodiments may only partially address some deficiencies or just one deficiency that may be discussed in the specification, and some embodiments may not address any of these deficiencies.

(19) Further, it will be understood that when an element is referred to as being “connected to” or “coupled to” another element, it may be directly connected to or coupled to the other element, or intervening elements may be present.

(20) In the present disclosure, not every layer of a circuit or package is depicted in the drawings, and some blocks may be omitted from the depicted circuit or package. One of ordinary skill in the art should understand that the circuit or package can include more layers to implement functionality

of the circuit or package and omitting these layers is merely for convenience of descriptions.

(21) The demand for higher memory capacity is incessant. Moreover, the higher the memory capacity, the higher the bandwidth of the memory circuit is demanded.

(22) A memory circuit may include a memory array of memory cells, each memory cell being capable of storing a certain amount of information, such as one or more bits. Usually, the memory array does not exist by itself but comes with other circuits that cooperate with the memory array, such as an address decoder for addressing one or more particular memory cells in the array and a sensing element for reading/writing information of the addressed memory cells.

(23) Circuit manufacturing technology, such as the technology for making semiconductor-based integrated circuits (IC), rapidly improves by a continuous decrease in the size of the smallest circuit element that can be made with a certain generation of the technology. For example, while a so-called 90-nanometer (nm) manufacturing process (sometimes referred to as the 90-nm technology node) was considered cutting-edge only several years ago, it is now possible to contemplate a 5-nm manufacturing process with mass production capability.

(24) Numerous semiconductor circuits of the same type can be made on one semiconductor wafer, which is then cut to produce tens or even hundreds of the semiconductor circuits of the same type, allowing for low-cost mass production. Each of the portion of the cut wafer containing one of the said type of circuit may be referred to as a “die,” as known by persons having ordinary skill in the art.

(25) Improvement in IC manufacturing technology allows for more circuit elements or devices to be made on one die of the same size. Hence, an improved IC manufacturing technology can result in a higher memory capacity in the same size of a die with memory circuits. An improved IC manufacturing technology can result in a higher computational power in the same size of a die with logic and/or computational circuits. Another way to phrase these improvements is the ability to provide more function/capacity into a smaller form factor, as known by persons having ordinary skill in the art.

(26) As mentioned before, a functional memory circuit may include a memory array of memory cells for storage and other circuits for other functions such as data retrieval, data writing, address decoding, logic and state control, data buffer, input/output (I/O) system and management, various kinds of drivers, I/O gating, and I/O interface with other hardware external to the memory circuit.

(27) Advancement in manufacturing technology that shrinks the size of individual memory cells is one of the main contributors to the growth in memory capacity. New applications, such as artificially intelligence (AI) applications, can often successfully utilize the higher memory capacity, and in turn generates more demand for it.

(28) Although the memory capacity can grow quickly, other performance metrics, such as bandwidth (the speed at which data can be read from/written into the memory circuit) and power consumption, do not necessarily improve at a proportional rate. One of the reasons may come from the limitation of the operation of memory cells that require a high voltage. Another reason may come from the performance requirement of the cell sensing circuits that may prevent full exploitation of the benefits of the latest manufacturing process. Also, the internal I/O system of some memory circuits has not migrated to the latest technology, thereby contributing to a less-than-optimal performance and higher power consumption. Hence, the benefits of the latest manufacturing process are sometime not fully exploited in memory circuits.

(29) Furthermore, the inventors recognized that some of the conventional ways of designing and making memory systems do not tend to allow easy customization of memory capacity and I/O width.

(30) Hence, there is a need for ways to better utilize the benefits of cutting-edge manufacturing technology regarding the designing and making of memory devices, circuits and systems.

(31) One of the advantages of the embodiments of the subject matter in the present disclosure is the ability to provide memory devices, circuits and systems with customizable memory capacity.

(32) One of the advantages of the embodiments of the subject matter in the present disclosure is the ability to provide memory devices, circuits and systems with customizable I/O width, and therefore better integration with other circuit blocks to form a larger, better-performing digital system.

(33) One of the advantages of the embodiments of the subject matter in the present disclosure is the ability to better exploit the benefits of improvements in manufacturing technology in the design and making of memory devices, circuits and systems. The benefits may include at least one of a higher memory capacity, a less power consumption, and a shorter access time, a higher memory bandwidth. Combinations of several or all aforementioned benefits can also be realized.

(34) To achieve at least one of the abovementioned technical advantages, the memory array of memory cells and the other circuits that do not provide the bulk of the storage capacity may be made on different dies and then combined together; the combination may be implemented in a plurality of ways. Some of such ways may relate to utilizing packaging techniques, such as 2.5D IC (e.g., the use of an interposer) and 3D IC (e.g., the use of die-to-die vias between two dies that are vertically displaced from each other).

(35) Manufacturing processes may thus be tailored to requirements that may be specific to the memory array and/or the other circuits (often logic circuits) that do not provide the bulk of the storage capacity. In other words, the manufacturing of the memory array does not have to be bound by considerations specific to the manufacturing of the other circuits that do not provide the bulk of the storage capacity, and vice versa.

(36) Independent designs of memory arrays and such other circuits (sometimes referred to as the “peripheral circuits”) thus become possible.

(37) A higher degree of customization in the communication interface between the two dies also becomes possible. In addition, a higher degree of customization in the communication interface with other hardware external to the memory circuit becomes possible.

(38) Furthermore, a reduction in power consumption by optimizing manufacturing processes for various individual circuit blocks becomes possible.

(39) Furthermore, implementing the bulk of the memory capacity and the associated logic circuits on separate dies can allow for flexible I/O density and/or pin density to each of the dies.

(40) Additional details of possible embodiments of the subject matter of the present disclosure are described below with the accompanying drawings.

(41) FIG. 1 illustrates a circuit **1** in accordance with some embodiments of the present disclosure. The circuit **1** is a memory circuit. The circuit **1** may be made by IC manufacturing technologies. The circuit **1** and/or some components thereof may be based on semi-conductive materials.

(42) The circuit **1** includes a die **10** and a die **11**. The die **10** may be referred to as a first die, and the die **11** may be referred to as a second die. The dies **10** and **11** may be coupled to each other by an interface **13**. The dies **10** and **11** may be based on semiconductors, and may be individually fabricated.

(43) The die **10** may include one or more memory arrays **100**. In the example of FIG. 1, the die **10** includes an array of memory arrays **100**, although this is not a limitation to the subject matter of the present disclosure. The size of the array is flexible, and may be determined based on the need for memory capacity. The memory array **100** includes memory cells and other circuits and will be described in more detail afterwards.

(44) In the exemplary array of FIG. 1, the number of rows may range from K to L, and the number of columns may range from P to Q, where K, L, P and Q are appropriate integers. In an embodiment, the designer of the circuit **1** and the die **10** does not have to start from scratch. Instead, the designer of the die **10** may choose from a library of well-defined circuits that may go into the die **10**. The library may be supplied by third-party intellectual property (IP) providers. The library may be supplied by specialists in IC manufacturing. Thus, the designer of the circuit **1** and the die **10** can flexibly choose the amount of memory capacity that the circuit **1** may need. This flexibility contributes the lowering of design cost. Using well-defined circuits from the library

supplied by IC manufacturing specialists may also contribute a reduction in the manufacturing cost because aspects of the circuits related to manufacturing have already been worked out and thought through by the IC manufacturing specialists.

(45) The die **11** may include various circuits, such as digital circuits and logic circuits, that can cooperate with the die **10** in order to make the circuit **1** a functional memory circuit, device and/or system. In an embodiment, most of the memory capacity of the circuit **1**, such as more than 90%, more than 95%, more than 97%, more than 99%, more than 99.9% and more than 99.99% of the memory capacity of the circuit **1**, is provided by the die **10**. However, this does not preclude the die **11** from having some memory capacity for the proper functioning of the circuits in the die **11**.

(46) In some embodiments, the die **11** may include blocks, circuits or modules such as logic controllers, state controllers, row latches, row decoders, column latches, column decoders, wordline driver pre-decoders, address decoders, write drivers, and circuits related to I/O between the dies **10** and **11** and to I/O between the circuit **1** and other circuits external to the circuit **1**. Hence, the die **11** may communicate with the die **10** to access the data stored in the memory arrays **100**. Access may include reading and writing. Hence, the die **11** may read data stored in some memory arrays **100** of the die **10** and output the data to circuits external to the circuit **1**. Hence, the die **11** may, upon request from circuits external to the circuit **1**, write data into some memory arrays **100** of the die **10**.

(47) The interface **13** may include direct electrical connection, and may be implemented with any appropriate ways, such as wires, pins and pads. Signals may be communicated via the interface **13**. Exemplary signals that may be transferred over the interface **13** include, but are not limited to, signals with routing information, signals related to I/O internal and/or external to the circuit **1**, signals related to column address control, signals related to row address control, signals related to data access, and signals related to timing and clock.

(48) It was mentioned that the dies **10** and **11** may be individually fabricated. Hence, in some embodiments, the dies **10** and **11** may be made from different manufacturing processes.

(49) Different manufacturing processes may be characterized in several ways. In some embodiments, a manufacturing process is characterized by the maximum degree of miniaturization of certain types of circuit elements that can be made by such process. For example, a manufacturing process may be characterized by the minimum channel length of the metal-oxide-semiconductor field-effect transistor (MOSFET) that can be fabricated by such process. The minimum channel length may be referred to as the “smallest feature size” in appropriate contexts known by persons having ordinary skill in the art. In some embodiments, the oxide thickness of the MOSFET may also contribute to the characterization of a manufacturing process. In some embodiments, a manufacturing process may be characterized by the so-called process node or technology node, which also describes the degree of miniaturization. For example, while the 10-micron process may have been considered advanced in the years of nineteen seventies, the 10-nanometer (10-nm) process is common today, representing a 1000-fold increase in the degree of miniaturization.

(50) In some embodiments, the die **11** is made with a manufacturing process that has a smallest feature size that is smaller than the manufacturing process associated with the fabrication of the die **10**. The other way round is also possible in some other embodiments. The smallest feature size may also be referred to as the critical dimension or the characteristic critical dimension because this dimension is a principal characteristic of a particular manufacturing process.

(51) The dies **10** and **11** may be distinguished from each other by ways, such as the maximum voltage or current at which some or all of the circuit blocks of the dies **10** and **11** can operate. For example, operating the circuits and/or dies at a higher maximum voltage may be suitable if a higher voltage swing is needed; on the other hand, operating the circuits and/or dies at a lower maximum voltage may be suitable for reducing power consumption and leakage.

(52) For example, the die **10** may require a larger voltage swing than the die **11** for having an

amplifier (such as sensing amplifier for reading the memory cells); too low an available voltage swing may reduce the performance of the sensing amplifier. In contrast, the die **11** may require a smaller voltage swing because most of the circuits in the die **11** are logical circuits that do not necessarily require a large dynamic range.

(53) Note that the available voltage swing on a die may be affected by various factors, one of them being the manufacturing process used to make such die.

(54) Hence, it can lead to further flexibility and optimization to use different manufacturing processes for making the die **10**, which provide the bulk of the memory capacity, and the die **11**, which does not provide the bulk of the memory capacity and mainly includes logic functionality of the memory circuit **1**.

(55) For example, the die **10** may be made by a process with a lower smallest feature size (e.g., 5 nm) compared to the die **11** (e.g., 10 nm), if a high memory density is required for the die **10**. In other situations, the die **11** may be made by a process with a lower smallest feature size compared to the die **10** if reducing the area of the die **11** is more important.

(56) For example, the memory die **10** may allow a higher voltage swing to improve the performance of the sensing operation, whereas the peripheral circuit die **11** may allow a lower voltage swing to improve power consumption.

(57) In some embodiments, the first die **10** may be referred to as the “memory” die. In some embodiments, the second die **11** may be referred to as the “peripheral circuit” die.

(58) FIG. 2 illustrates a memory array **100** in accordance with some embodiments of the present disclosure.

(59) The memory array **100** includes memory cells **101**, a sensing element **103**, a wordline driver **105** or a portion thereof, and at least one electrical connection **107**.

(60) The memory cells **101** are capable of storing information and may be implemented with various kinds of appropriate data storage devices or state storage devices. A data storage device that stores one bit, which is the smallest unit of information, can sometimes be referred to as a bit storage device. The operation of data storage device may be based on the storage of electric charges, electric voltages, electric resistances, magnetic states, ferromagnetic states, optical states and/or other appropriate mechanisms.

(61) As the plurality of memory arrays **100** in the die **10** may be organized by rows and columns, the memory cells **101** may also be organized by rows and columns. The combination of a specific row and a specific column may allow the addressing of one or more specific data storage devices and thus may be referred to as an address. The addressing may be performed by an address decoder, which, in some embodiments, may reside on the second die **11**. In some embodiments, the rows of the memory cells **101** may be accessed by word lines. In some embodiments, the columns of the memory cells **101** may be accessed by bit lines.

(62) The sensing element **103** is used to sense the state of the memory cells **101**, thereby allowing the reading of data stored in the memory cells **101**. In some embodiments, the sensing element **103** includes a sense amplifier to amplify the electric voltage stored in the particular memory cell that is being read. The sense amplifier may be implemented with electronic circuitry.

(63) The wordline driver **105** may help access rows of the memory cells **101**. The wordline driver **105** may help write data into the memory cells **101**. In some embodiments, only a portion of the wordline driver **105** is included in the memory array **100**; in such a case, the other portions of the wordline driver may reside in the second die **11**. In an embodiment, only the wordline driver buffer is included in block **105** within the memory array **100**.

(64) The electrical connections **107** may take many forms, such as wires, pins and pads. The electrical connections **107** connect part or all of the circuitry in the memory array **100** to circuitry outside the memory array **100**.

(65) FIG. 3 illustrates a circuit **2** in accordance with some embodiments of the present disclosure.

(66) The circuit **2** includes a die **20** and a die **21**, which may be respectively referred to as the first

die **20** and the second die **21**. Each of the dies **20** and **21** may include several circuit blocks or modules.

(67) The die **20** illustrated in the embodiment of FIG. **3** includes a plurality of memory arrays **200**. Each of the memory arrays **200** may include memory cells **201**, a sensing element **203**, a wordline driver buffer **205**, and electrical connections **207**. The memory array **200** in FIG. **3** may be similar to the memory array **100** in FIG. **2**, so a detailed description of the memory array **200** can be omitted here.

(68) The die **21** illustrated in the embodiment of FIG. **3** includes several blocks **211**, **212**, **213**, **214**, **215** and **216**, each of which may serve a particular function. Note the functions illustrated in the blocks of FIG. **3** are merely exemplary and are not intended as limitation.

(69) The block **211** may include control logic suitable for controlling the operation of the die **21** in terms of at least its interaction with the die **20** and external circuits and systems. The block **211** may receive various signals, such as clock signals, address signals and data signals, and may instruct other blocks in the die **21** to interact with the die **20**.

(70) Blocks **212** and **213** constitute an address decoder suitable for addressing particular memory cells **201** in the memory array **200** of the memory die **20**. In some embodiments, block **212** may include a wordline driver pre-decoder.

(71) Block **214** may include write drivers and circuits/modules related to I/O interfaces with the die **20**. In an embodiment, the die **21** is connected to the sensing element **203** of the die **20**. In an embodiment, the die **21** is directed connected to the sensing element **203** of the die **20** through an I/O block of the die **21**.

(72) Block **214** may be used during a write operation (through, e.g., I/O circuitry therein) to write data into the memory cells **201**. Block **214** may be used during a read operation to read out data output by the sensing element **203** of the die **20**.

(73) Data that the block **214** receives from the sensing element **203** during a read operation may be first stored in a data buffer before being transferred to circuits external to the circuit **2**. Data that are received from circuits external to the circuit **2** for being written into the memory array **200** of the die **20** may arrive a data buffer first before reaching block **214** and eventually being written into the memory array **200**. Hence, block **215** may be an data in/out buffer.

(74) Block **216** may represent the actual physical features that implement the I/O functions regarding circuits external to the circuit **2**, such as pads, pins, conductive posts and pillars, etc. Data being read from or to be written to the memory arrays **200** may go through block **216**.

(75) FIG. **4** illustrates a memory circuit **3** in accordance with some embodiments of the present disclosure.

(76) The memory circuit **3** illustrated in FIG. **4** includes a first die **30** and a second die **31**. The die **30** includes blocks **301** and **303**, and the die **31** includes blocks **311** and **313**. It is understood that the dies **30** and **31** may include other circuits blocks and modules.

(77) Block **301** schematically illustrates some of the memory cells that provide most of the memory capacity of the circuit **3**. An individual memory cell in the embodiment of FIG. **4** includes a switch (which may be implemented with a transistor) and a bit storage device (which may be implemented with a capacitor). The memory cells may be accessed by bit lines BL and complementary bit lines BLB.

(78) Block **303** schematically illustrates circuitry for accessing the memory cells in block **301**. Included in block **303** of the embodiment of FIG. **4** are a sense amplifier that can be controlled by a signal SAEN and its complement SAENB for controlling whether the sense amplifier is enabled or not. Block **303** may also include circuitry controlled by the signal CS (cell selection) for controlling the selection of particular cells to be accessed.

(79) Block **311** schematically illustrates a portion of the write driver, which can be controlled by the signal WE (write enable) and a pair complementary input signals IN and INB.

(80) Block **313** schematically illustrates other circuit blocks that do not provide the bulk of memory

capacity of the circuit 3 but may be useful for accessing and/or interacting with the die 30 and circuits external to the circuit 3.

(81) FIG. 5 illustrates a memory circuit 4 in accordance with some embodiments of the present disclosure.

(82) The circuit 4 shown in FIG. 5 includes a first die 40 comprising arrays of memory cells, a second die 41 comprising other circuits suitable for interacting with the die 40 and external circuits, and an interposer 43 that couples the die 40 to the die 41. Each of the dies 40 and 41 may include electrical connections 47 such as pins, conductive posts and conductive pillars. The dies 40 and 41 may be separated from each other by a gap 45.

(83) The dies 40 and 41 and the interposer 43 may, together with other appropriate elements not shown in FIG. 5, form an electronic package. For example, an encapsulant may be used to further hold the die 40, the die 41 and the interposer 43 together. The encapsulant may include a molding compound (such as epoxy) and/or a wafer-level underfill (i.e., electrically insulating adhesive material applied to the die 40 and/or the die 41, for example during wafer-level processes).

(84) The encapsulant may partially or fully fill the gap 45. The encapsulant may encapsulate some of the electrical connections 47 of the dies 40 and 41. In some embodiments, the encapsulant may fully encapsulate all of the electrical connections 47 of the die 40, fully encapsulate all of the electrical connections 47 of the die 41 or both.

(85) In some embodiments, the encapsulant may partially encapsulate the die 40, the die 41 and/or the interposer 43. In some embodiments, the encapsulant may encapsulate the entirety of the die 40, the die 41, and/or the interposer 43.

(86) The memory circuit 4 shown in FIG. 5 may be made by 2.5D IC packaging technology.

(87) The dies 40 and 41 may be distinguished from each other by the types of the electrical connections 47 thereof. For example, an electrical connection 47 with a lower current rating may have a smaller volume and thus suitable for contributing the smaller form factor. On the other hand, an electrical connection 47 with a higher current rating if the die it is connected to requires a higher current.

(88) Although only one die 40 and one die 41 are illustrated in FIG. 5, the number of dies are not limited. For example, it is possible for the memory circuit to have one die 41 that works with more than one memory dies 40; in which case, each of the memory dies 40 may be made by different manufacturing processes and/or have different electrical characteristics and/or pin and I/O densities.

(89) FIG. 6 illustrates a memory circuit 5 in accordance with some embodiments of the present disclosure.

(90) The circuit 5 shown in FIG. 6 includes a first die 50 comprising arrays of memory cells, and a second die 51 comprising other circuits suitable for interacting with the die 50 and external circuits. The dies 50 and 51 may be connected to each other by way of electrical connections 53, such as vias, die-to-die vias, through-silicon vias (TSV) and conductive pillars. Gap 55 may exist between the electrical connections 53; gap may be partially or fully filled with encapsulant.

(91) The dies 50 and 51 and the electrical connections 53 may, together with other appropriate elements not shown in FIG. 6, form an electronic package. For example, an encapsulant may be used to strengthen the die 50, the die 51 and the electrical connections 53 together. The encapsulant may include a molding compound (such as epoxy) and/or a wafer-level underfill (i.e., electrically insulating adhesive material applied to the die 50 and/or the die 51, for example during wafer-level processes).

(92) The encapsulant may encapsulate some of the electrical connections 53 of the dies 50 and 51. In some embodiments, the encapsulant may fully encapsulate all of the electrical connections 53 of the die 50, fully encapsulate all of the electrical connections 53 of the die 51 or both.

(93) In some embodiments, the encapsulant may partially encapsulate the die 50, the die 51 and/or the electrical connections 53. In some embodiments, the encapsulant may encapsulate the entirety

of the die **50**, the die **51**, and/or the electrical connections **53**.

(94) The dies **50** and **51** may be formed on different levels. The die **50** may be above the die **51**, and vice versa.

(95) The memory circuit **5** shown in FIG. **6** may be made by 3D IC packaging technology.

(96) Although only one die **50** and one die **51** are illustrated in FIG. **6**, the number of dies are not limited. For example, it is possible for the memory circuit to have one die **51** that works with more than one memory dies **50**; in which case, each of the memory dies **50** may be made by different manufacturing processes and/or have different electrical characteristics and/or pin and I/O densities.

(97) FIG. **7** illustrates an exemplary flowchart in accordance with some embodiments of the present disclosure.

(98) At step **701**, a first die is provided. The first die may include a memory array comprising a plurality of memory cells and a sensing element. At step **703**, a second die is provided. The second die may include an address decoder associated with the memory array of the first die.

(99) At step **705**, the second die is coupled to the sensing element of the first die. At step **707**, an encapsulant that at least partially encapsulates the first die and the second die is provided.

(100) FIG. **8** illustrates an exemplary flowchart in accordance with some embodiments of the present disclosure.

(101) At step **801**, a first die is provided. The first die may include a memory array comprising a plurality of memory cells, a sensing element, and a first plurality of conductive pads coupled to the sensing element. At step **803**, a second die is provided. The second die may include an address decoder associated with the memory array of the first die and a second plurality of conductive pads coupled to the address decoder.

(102) At step **805**, the first plurality of conductive pads are coupled to the second plurality of conductive pads. The first plurality of conductive pads may be operable at a first maximum voltage. The second plurality of conductive pads may be operable at a second maximum voltage higher than the first maximum voltage.

(103) FIG. **9** is a block diagram of an integrated circuit (IC) manufacturing system **1000**, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some embodiments, based on a layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using manufacturing system **1000**.

(104) In FIG. **9**, IC manufacturing system **1000** includes entities, such as a design house **1020**, a mask house **1030**, and an IC manufacturer/fabricator ("fab") **1050**, that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **1060**. The entities in system **1000** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **1020**, mask house **1030**, and IC fab **1050** is owned by a single larger company. In some embodiments, two or more of design house **1020**, mask house **1030**, and IC fab **1050** coexist in a common facility and use common resources.

(105) Design house (or design team) **1020** generates an IC design layout diagram **1022**. IC design layout diagram **1022** includes various geometrical patterns designed for an IC device **1060**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **1060** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **1022** includes various IC features, such as an active region, gate electrode, source and drain, metal lines or vias of an

interlayer interconnection, and openings for bonding pads, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **1020** implements a proper design procedure to form IC design layout diagram **1022**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **1022** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **1022** can be expressed in a GDSII file format or DFII file format.

(106) Mask house **1030** includes data preparation **1032** and mask fabrication **1044**. Mask house **1030** uses IC design layout diagram **1022** to manufacture one or more masks **1045** to be used for fabricating the various layers of IC device **1060** according to IC design layout diagram **1022**. Mask house **1030** performs mask data preparation **1032**, where IC design layout diagram **1022** is translated into a representative data file (“RDF”). Mask data preparation **1032** provides the RDF to mask fabrication **1044**. Mask fabrication **1044** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **1045** or a semiconductor wafer **1053**. The design layout diagram **1022** is manipulated by mask data preparation **1032** to comply with particular characteristics of the mask writer and/or requirements of IC fab **1050**. In FIG. 9, mask data preparation **1032** and mask fabrication **1044** are illustrated as separate elements. In some embodiments, mask data preparation **1032** and mask fabrication **1044** can be collectively referred to as mask data preparation.

(107) In some embodiments, mask data preparation **1032** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **1022**. In some embodiments, mask data preparation **1032** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

(108) In some embodiments, mask data preparation **1032** includes a mask rule checker (MRC) that checks the IC design layout diagram **1022** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some embodiments, the MRC modifies the IC design layout diagram **1022** to compensate for limitations during mask fabrication **1044**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

(109) In some embodiments, mask data preparation **1032** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **1050** to fabricate IC device **1060**. LPC simulates this processing based on IC design layout diagram **1022** to create a simulated manufactured device, such as IC device **1060**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (“DOF”), mask error enhancement factor (“MEEF”), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **1022**.

(110) It should be understood that the above description of mask data preparation **1032** has been simplified for the purposes of clarity. In some embodiments, data preparation **1032** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **1022** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **1022** during data preparation **1032** may be executed in a variety of different orders.

(111) After mask data preparation **1032** and during mask fabrication **1044**, a mask **1045** or a group of masks **1045** are fabricated based on the modified IC design layout diagram **1022**. In some embodiments, mask fabrication **1044** includes performing one or more lithographic exposures based on IC design layout diagram **1022**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **1045** based on the modified IC design layout diagram **1022**. Mask **1045** can be formed in various technologies. In some embodiments, mask **1045** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **1045** includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the binary mask. In another example, mask **1045** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **1045**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **1044** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **1053**, in an etching process to form various etching regions in semiconductor wafer **1053**, and/or in other suitable processes.

(112) IC fab **1050** includes wafer fabrication **1052**. IC fab **1050** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **1050** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

(113) IC fab **1050** uses mask(s) **1045** fabricated by mask house **1030** to fabricate IC device **1060**. Thus, IC fab **1050** at least indirectly uses IC design layout diagram **1022** to fabricate IC device **1060**. In some embodiments, semiconductor wafer **1053** is fabricated by IC fab **1050** using mask(s) **1045** to form IC device **1060**. In some embodiments, the IC fabrication includes performing one or more lithographic exposures based at least indirectly on IC design layout diagram **1022**.

Semiconductor wafer **1053** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **1053** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

(114) In the present disclosure, memory circuits and related methods with improvement are disclosed. The memory circuit may include multiple dies. The bulk of the memory capacity may be provided in one or more of the multiple dies. The related logic and/or I/O functions may be provided in others of the multiple dies. The multiple dies may be packaged together. These configurations can allow for optimization in manufacturing processes, electrical characteristics, and I/O and pin densities among different dies with different functions.

(115) It will be understood that not all advantages have been necessarily discussed herein, no particular advantage is required for all embodiments or examples, and other embodiments or examples may offer different advantages.

(116) According to an aspect of the present disclosure, a circuit is provided. The circuit includes a first die that includes a memory array, and the memory array includes a plurality of memory cells, a sensing element coupled to the plurality of memory cells, and a first plurality of conductive pads coupled to the sensing element. The circuit also includes a second die that includes an address decoder associated with the memory array of the first die and a second plurality of conductive pads

coupled to the address decoder. The first die is coupled to the second die by an interposer. The address decoder of the second die is coupled to the sensing element of the first die. A first voltage swing of the first die is larger than a second voltage swing of the second die.

(117) In an embodiment, the first die may include a first plurality of conductive posts coupled to the sensing element of the first die. The second die may include a second plurality of conductive posts. The second die may be coupled to the sensing element of the first die by coupling the first plurality of conductive posts to the second plurality of conductive posts.

(118) In an embodiment, the interposer couples the first plurality of conductive posts to the second plurality of conductive posts.

(119) According to an aspect of the present disclosure, a circuit is provided. The circuit may include a first die that includes a memory array, and the memory array may include a plurality of memory cells, a sensing element coupled to the plurality of memory cells, and a first plurality of conductive pads coupled to the sensing element. The circuit may include a second die that includes an address decoder associated with the memory array of the first die and a second plurality of conductive pads coupled to the address decoder. The first die may be coupled to the second die by die-to-die vias. The address decoder of the second die may be coupled to the sensing element of the first die. A first voltage swing of the first die is larger than a second voltage swing of the second die.

(120) According to an aspect of the present disclosure, a circuit is provided. The circuit may include a first die that includes a plurality of memory arrays each including a plurality of memory cells and a sensing element coupled to the plurality of memory cells. A first smallest feature size may be associated with the first die. The circuit may include a second die including an address decoder associated with the memory array of the first die. A second smallest feature size that is smaller than the first smallest feature size may be associated with the second die. The address decoder of the second die may be coupled to the sensing elements of the first die. The plurality of memory arrays may be organized by rows and columns.

(121) In an embodiment, the sensing element may be configured to output data stored in the plurality of memory cells of the first die to the second die during a read operation of the memory circuit. In an embodiment, the second die may comprise a system I/O circuit configured to, during the read operation, receive the data output from the sensing element of the first die.

(122) In an embodiment, the second die may include at least one from the group consisting of: a logic controller, a row latch, a row decoder, a column latch, a column decoder, and a wordline driver pre-decoder.

(123) In an embodiment, the first die may be coupled to the second die by an interposer. In an embodiment, the encapsulant may at least partially encapsulate the interposer.

(124) In an embodiment, the first die may be coupled to the second die by die-to-die vias. In an embodiment, the encapsulant may at least partially encapsulate at least some of the die-to-die vias. In an embodiment, the encapsulant may fully encapsulate all of the die-to-die vias.

(125) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A circuit, comprising: a first die comprising a memory array, the memory array comprising: a plurality of memory cells; a strip of sensing elements arranged at a first side of the plurality of memory cells and coupled to the plurality of memory cells; and a strip of wordline drivers arranged at a second side of the plurality of memory cells and perpendicular to the strip of sensing elements, the strip of wordline drivers coupled to the plurality of memory cells; a second die comprising an address decoder associated with the memory array of the first die; and an interposer coupling the first die to the second die, wherein the first die and the second die are disposed on a side of the interposer, wherein a first voltage swing of the first die is different from a second voltage swing of the second die.
2. The circuit of claim 1, further comprising: an encapsulant at least partially encapsulating the first die and the second die.
3. The circuit of claim 2, wherein the encapsulant is a molding compound or a wafer-level underfill material.
4. The circuit of claim 1, wherein the first die comprises a first plurality of conductive posts coupled to the strip of sensing elements of the first die, wherein the second die comprises a second plurality of conductive posts coupled to the first plurality of conductive posts.
5. The circuit of claim 4, wherein an encapsulant at least partially encapsulates the first plurality of conductive posts and the second plurality of conductive posts.
6. The circuit of claim 1, wherein the first voltage swing of the first die is larger than the second voltage swing of the second die.
7. The circuit of claim 1, wherein the first die is made from a first manufacturing process, wherein the second die is made from a second manufacturing process, wherein a technology node of the second manufacturing process differs from a technology node of the first manufacturing process.
8. The circuit of claim 7, wherein the technology node of the second manufacturing process is smaller than the technology node of the first manufacturing process.
9. A circuit, comprising: a first die comprising a memory array, the memory array comprising: a plurality of memory cells; a strip of sensing elements arranged at a first edge of the plurality of memory cells and coupled to the plurality of memory cells; and a strip of wordline drivers arranged at a second edge of the plurality of memory cells and perpendicular to the strip of sensing elements, the strip of wordline drivers coupled to the plurality of memory cells; and a second die comprising an address decoder associated with the memory array of the first die; wherein the first die is coupled to the second die by die-to-die vias, and wherein a first voltage swing of the first die is different from a second voltage swing of the second die, and wherein the first die is made from a first manufacturing process, and the second die is made from a second manufacturing process, wherein a first technology node of the first manufacturing process differs from a second technology node of the second manufacturing process.
10. The circuit of claim 9, wherein the strip of sensing elements is configured to output data stored in the plurality of memory cells of the first die to the second die during a read operation.
11. The circuit of claim 10, wherein the second die comprises a system I/O circuit coupled to the address decoder through a first plurality of conductive pads and configured to, during the read operation, receive the data output from the strip of sensing elements of the first die.
12. The circuit of claim 11, wherein the system I/O circuit is configured to, during the read operation, output the data received by the system I/O circuit from the first die to outside the second die by way of at least some of the first plurality of conductive pads.
13. The circuit of claim 9, wherein the first die comprises a first plurality of conductive pillars coupled to the strip of sensing elements, wherein the second die comprises a second plurality of conductive pillars coupled to the first plurality of conductive pillars, wherein a first electric current rating is associated with the first plurality of conductive pillars and a second electric current rating higher than the first electric current rating is associated with the second plurality of conductive

pillars.

14. The circuit of claim 9, further comprising an encapsulant that at least partially encapsulates at least some of the die-to-die vias.

15. A circuit, comprising: a first die comprising a plurality of memory arrays each comprising: a plurality of memory cells; a strip of sensing elements arranged at a first edge of the plurality of memory cells and coupled to the plurality of memory cells; and a strip of wordline drivers arranged at a second edge of the plurality of memory cells and perpendicular to the strip of sensing elements, the strip of wordline drivers coupled to the plurality of memory cells, wherein a first smallest feature size is associated with the first die; and a second die comprising an address decoder associated with the memory array of the first die, wherein a second smallest feature size that is smaller than the first smallest feature size is associated with the second die; wherein the address decoder of the second die is coupled to the strip of sensing elements of the first die.

16. The circuit of claim 15, wherein the address decoder of the second die is configured to access a particular memory cell of the plurality of memory arrays of the first die by an address of the particular memory cell.

17. The circuit of claim 15, wherein the first die is coupled to the second die by an interposer.

18. The circuit of claim 15, wherein the first die is coupled to the second die by die-to-die vias.

19. The circuit of claim 15, wherein the second die comprises a write driver configured to, during a write operation, write data to the memory cells of the plurality of memory arrays of the first die.

20. The circuit of claim 15, wherein the first die comprises a first plurality of conductive posts coupled to the strip of sensing elements of the first die, wherein the second die comprises a second plurality of conductive posts coupled to the first plurality of conductive posts.
